



Bambusae Caulis in Taeniam extract reduces ovalbumin-induced airway inflammation and T helper 2 responses in mice

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ABSTRACT

Ethnopharmacological relevance: Bambusae Caulis in Taeniam (BC) was known as traditional herbal medicine with anti-inflammatory property in the Orient.

Aim of the study: Allergic asthma is inflammatory disease of airways associated with enhanced T helper (Th) 2 lymphocytes responses to allergens, leading to eosinophilic infiltration and elevated serum IgE levels. Although there were some studies that BC extract had an anti-inflammatory property, there was no study on asthma. In present study, we investigated the suitability of BC extract as a therapeutic candidate in the treatment of allergic airway disease in ovalbumin-induced asthma model.

Materials and methods: Balb/C mice (female, 6 weeks old) were treated by ovalbumin sensitization and nebulization, and used as asthma model. The number of eosinophil in bronchoalveolar lavage (BAL) fluid and the degree of eosinophilia were investigated by hematoxylin and eosin stain and the infiltration of inflammatory cells into lung tissues was examined by staining by hematoxylin and eosin solution. The levels of interleukin (IL)-4 in BAL fluid, immunoglobulin E (IgE) in serum, interferon (IFN)- γ and IL-4 production in splenocyte culture from Balb/C mice (not treated, 6 weeks old) that incubated with or without BC extract for 48 h were determined by enzyme-linked immunosorbent assay.

Results: The level of eosinophils was decreased by treatment of the animals with BC extract (40 mg/kg) and correspondingly, a significantly lowered degree of eosinophilia was also reported ($p < 0.01$). In lung tissue, BC extract reduced the increased immune cell infiltration induced by OVA ($p < 0.05$). Furthermore, the levels of IL-4 and IgE in BAL fluid or serum up-regulated by OVA was decreased by BC extract. Finally, IFN- γ production was significantly increased ($p < 0.01$), while IL-4 production significantly decreased ($p < 0.01$), after treatment of the culture supernatants of splenocytes with BC extract.

Conclusions: These results indicated that BC extract reduces OVA-induced airway inflammation and Th 2 response in mice, suggesting that BC extract can be a therapeutic candidate for allergic airway disease, including asthma.

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1. Introduction

Asthma is one of the most common chronic respiratory diseases in industrialized society and has emerged as a worldwide public health problem. Although many patients were received with conventional medical treatment for asthma, effects of therapy are frequently poor (Bender et al., 2007). In other side, the interest of complementary and alternative medicine (CAM) for asthma is increasing more and more (Engler, 2000; Bielory et al., 2004; Bukutu et al., 2008), and herbal medicine is one of the most popular CAM among asthma patients (Huntley and Ernst, 2000).

Allergic asthma is inflammatory disease of airways associated with enhanced T helper (Th) 2 lymphocytes responses to allergens, leading to eosinophilic infiltration, airway hyperresponsiveness, reversible airway obstruction, and elevated serum IgE levels (Umetsu et al., 2002). Interleukin-4 (IL-4), an attractive Th 2 cytokine increased in asthma, increase IgE levels (Haas et al., 1999) and the increase of IL-4 plays important roles in phenomenon of asthma such as above that (Woodfolk, 2006). Moreover, interferon (IFN)- γ regulates eosinophil recruitment in airway, so that IFN- γ , regulated Th 2 response, also was important in asthma (Iwamoto et al., 1993). Therefore, the regulation of Th 2 type lymphocytes responses by IFN- γ can be a target in allergic asthma.

Bambusae Caulis in Taeniam (BC) is the stem of *Phyllostachys nigra* var. *henosis* or *Phyllostachys bambusoides* (Family: Poaceae). BC is obtained by following steps: scraping off the coat from the

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bamboo stem, cutting the stem to slices, and binding them together followed by drying in shaded places. BC was known as traditional herbal medicine with anti-inflammatory property in the Orient (Shibata et al., 1975). Anti-oxidative properties of BC were reported by recent studies (Jiao et al., 2007; Jung et al., 2007). However, there is no study on effects of BC in ovalbumin (OVA)-induced asthma model. Thus, the goal of this study was to evaluate the ethnopharmacological efficacy of BC in OVA-induced asthma model.

In present study, we found out that BC extract inhibited the infiltration of eosinophils, and decreased the expression of IL-4 in bronchoalveolar lavage (BAL) fluid and IgE in serum on ovalbumin (OVA)-induced asthma model. Furthermore, BC extract enhanced Th differentiation into Th-1 lymphocytes in splenocytes culture. Thus, BC extract can be a therapeutic candidate for allergic airway disease.

2. Materials and methods

2.1. Preparation of the BC and *P. nigra* var. *henosis* extracts

BC extract was purchased from the Sun Ten Pharmaceutical (Taipei, Taiwan). The BC extract contains the extract of BC (67%) and corn starch (33%). The BC extract was weighed precisely to 298.5 mg (contain 200 mg extract of BC) and was dissolved in 20 ml distilled water (DW) using stirrer, overnight, at room temperature. For starch-separation, the dissolved solution was centrifuged for 30 min at 3000 rpm and 4 °C. The supernatant was sterilized by passing through 0.22 µm syringe filter (Millipore, MA, USA). Then the stock of BC extract was diluted in several concentrations and used in the experiments. The stock (KHOP00209) was stored at 4 °C before using. The middle layer of *P. nigra* var. *henosis* was purchased from Omniherb (Daegu, Korea) and the identity was confirmed by one of the authors (Y.P. Jang). The voucher specimen (KHOP00315) was secured at Kyung Hee Korean Traditional Herbal Medicine Museum of College of Pharmacy, Kyung Hee University. *P. nigra* var. *henosis* (360 g) was extracted with 5 L of water two times at 100 °C for 2 h. After filtration, water extract was lyophilized with freezing dryer to give the solid *P. nigra* var. *henosis* extract (28 g). The *P. nigra* var. *henosis* extract was dissolved 10 mg/ml in DW and filtered by 0.45 µm syringe filter (Millipore, MA, USA).

2.2. Induction of asthma in Balb/C mice and drug administration

Six-week-old female Balb/C mice were purchased from Orient Bio Inc. (Korea). They were housed under specific pathogen free conditions (about 22 ± 1 °C). All animal study protocols were reviewed and approved by the Committee for the Care and Use of Laboratory Animals (College of Pharmacy, Kyung Hee University). In a preliminary dose-response experiment, the amount of BC extract was assessed for detection of anti-inflammatory effects. The concentration of BC extract was decided at 40 mg/kg, the minimum dosage with effective (data not shown). In subsequent experiment, mice were treated with OVA (Grade V; Sigma–Aldrich, MO, USA) with a slightly modified version of the standardized procedure, consisting of a first phase of sensitization and a second phase of induction of allergic response (Gonzalo et al., 1996). Thus, mice were sensitized with OVA (0.1 mg/mouse) with 10 mg aluminium hydroxide (Sigma–Aldrich, MO, USA) as an adjuvant via intraperitoneal on day 1 followed by exposure to aerosolized antigen (5% OVA for 30 min on day 22 and 1% OVA for 30 min on days 29, 31 and 33) using an ultrasonic nebulizer (ME-U12, Omron, Japan) to induced the response. The mice were divided into four groups ($n=5$), and airway inflammation was induced by OVA in three groups using method described in Fig. 1. Among two groups of asthma-induced mice, one was treated with 1 mg/kg of dexametha-

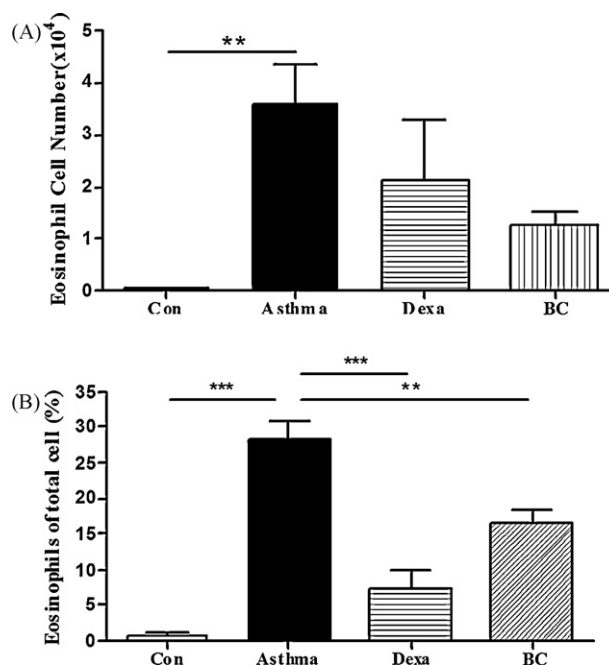


Fig. 1. Effect of BC extract on the recruitment of inflammatory cells in BAL fluid obtained from OVA-induced asthma model. The OVA-induced asthma model was divided into 3 treatment groups; untreated (Asthma), treated with 1 mg/kg of dexamethasone (Dexa), or treated with 40 mg/kg of BC extract (BC) for inhalation periods. The negative control group (Con) was sensitized and treated with PBS without drug administration. The number of eosinophils in the fluid was counted after hematoxylin and eosin stain (A and B). The results were expressed as the means ± S.E.M. ** $p < 0.01$ and *** $p < 0.001$ were considered indicative of statistically significant differences between the control group and asthma group or between the asthma group and medicine-treated group.

sone (Dexa) (Sigma–Aldrich, MO, USA) and the other was treated with 40 mg/kg of BC extract between days 28 and 34 via orogastric gavage. The control group was sensitized and challenged with phosphate buffered saline (PBS) without treatment via orogastric gavage. Mice were sacrificed on day 35.

2.3. Collection of BAL fluid

BAL fluid was collected by the lung lavage with 0.9 ml of PBS via the trachea. After three lavages, approximately 0.7 ml of BAL fluid was recovered and centrifuged for 10 min at 400 × g and 4 °C. The supernatant was stored at –70 °C for the measurement of cytokines. The cells in the BAL fluid were resuspended in 100 µl of PBS for total cell and differential counts. After the measurement of total cell number using a neubauer hemocytometer (Hausser Scientific, PA, USA), the cells suspended in PBS were applied to a slide by cytopinning and performed the hematoxylin and eosin stain. The percentage of each leukocyte was determined, and the absolute leukocyte count was calculated by multiplying each leukocyte percentage by the total cell count.

2.4. Histological analysis

Lung tissues were detached from the mice and fixed with 4% formaldehyde for overnight at 4 °C. The fixed tissues were embedded in paraffin and cut into 4 µm sections with a microtome (Leica, Nussloch, Germany). The sections were placed on slide glasses, deparaffinized, and stained with hematoxylin and eosin (Sigma–Aldrich, MO, USA) in order to examine the cells that had infiltrated into the peribronchial connective tissues. A peribronchial cell count based on a five-point scoring system was performed as previously described to estimate the severity of

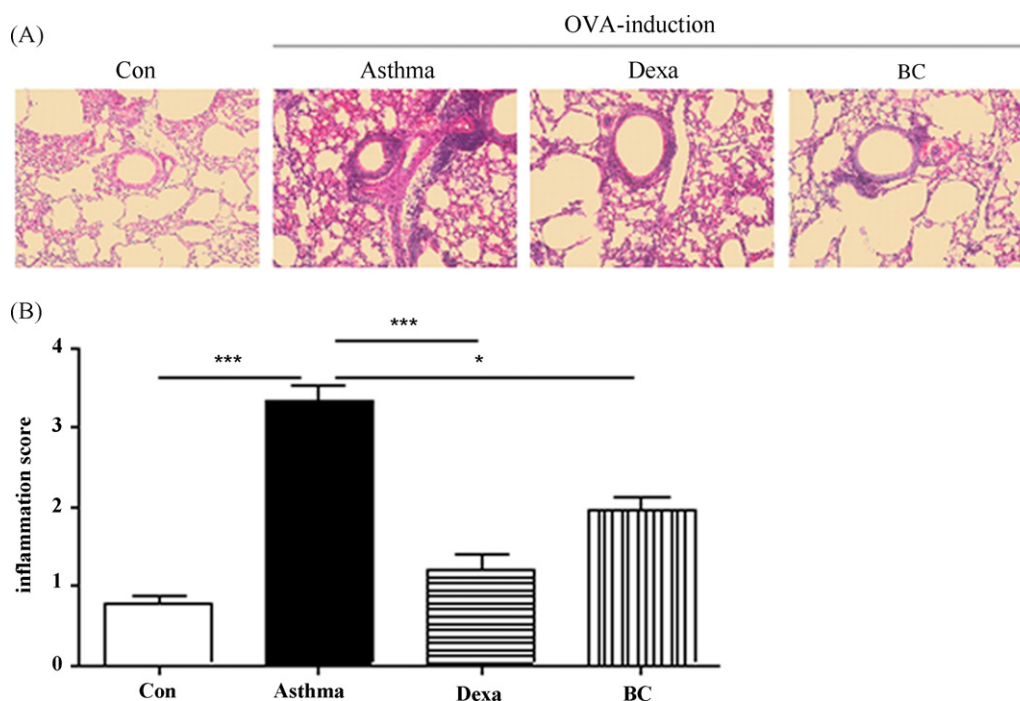


Fig. 2. Effect of BC extract on airway inflammation caused by cell infiltration in lung tissue. To confirm the pathologic changes in lung tissues, we performed hematoxylin and eosin staining (A) and scored the inflammatory cell in filtration in the lung tissues (B) as described in method. Con, sensitized and treated with PBS without drug administration; Asthma, OVA-induced asthma model with untreated; Dexa, OVA-induced asthma model treated with 1 mg/kg of dexamethasone; BC, OVA-induced asthma model treated with 40 mg/kg of BC extract. Magnification 200 $\times$.

leukocyte infiltration (Duan et al., 2004). The scoring system was: 0, no cells; 1, a few cells; 2, a ring of cells 1 cell layer deep; 3, a ring of cells 2–4 cell layers deep; and 4, a ring of cells more than 4 cell layers deep. The leukocyte scoring was examined in three independent fields of lung section from each mouse.

2.5. Measurement of cytokine

The concentrations of IFN- γ , IL-4 and IgE were measured by a sandwich ELISA using Mouse CytoSet Kit (Invitrogen, CA, USA) and OptEIA Set mouse IgE (BD Biosciences, CA, USA). All assays were performed in triplicate. The concentration of each protein was calculated from the standard curve.

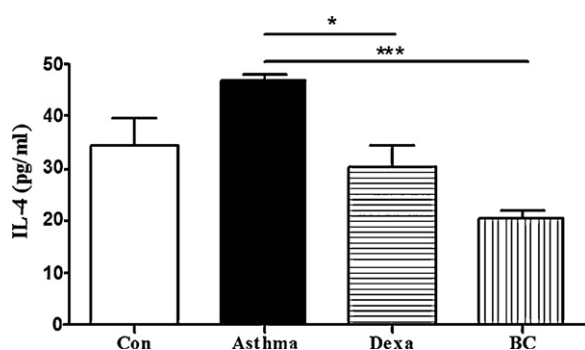


Fig. 3. Effect of BC extract on the expression of IL-4 in BAL fluid. The BAL fluid was collected after the cells in the fluid were eliminated. The protein expression of IL-4 in BAL fluid was analyzed by ELISA. The data are expressed as the means $\pm$ S.E.M. * p < 0.05 and *** p < 0.001 were considered indicative of statistically significant differences between the asthma group and medicine-treated group. Con, sensitized and treated with PBS without drug administration; Asthma, OVA-induced asthma model with untreated; Dexa, OVA-induced asthma model treated with 1 mg/kg of dexamethasone; BC, OVA-induced asthma model treated with 40 mg/kg of BC extract.

2.6. Preparation of splenocytes

All procedures were conducted under aseptic conditions. Six-week-old female Balb/C mice were killed by cervical dislocation and a single cell suspension was prepared by pressing the spleen between two slide glasses. The cell suspensions were passed through a 70 μ m cell strainer (BD Falcon, MA, USA) and were centrifuged at 1500 rpm for 5 min. The supernatant was removed. After tapping, cells incubated with 0.75 ml of RBC lysis buffer (Sigma–Aldrich, MO, USA) for 1 min and were washed three times in RPMI 1640, and resuspended and adjusted to 2×10^6 cells/ml with RPMI (Hyclone, UT, USA) supplemented with 10% heat-inactivated fetal bovine serum (Hyclone, UT, USA), 100 U/ml penicillin, 100 μ g/ml streptomycin, 2 mM L-glutamine

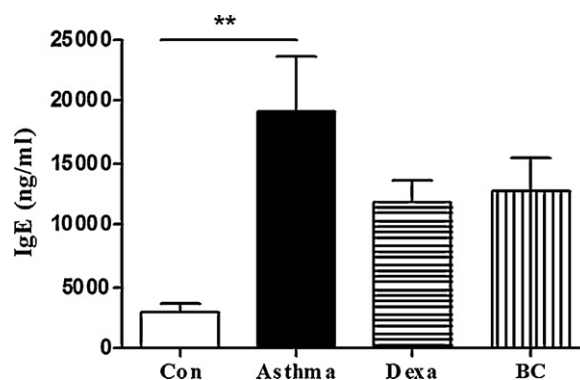


Fig. 4. Effect of BC extract on total IgE level in the serum. The total IgE levels in the serum were analyzed by ELISA as described in Section 2. The data are expressed as the means $\pm$ S.E.M. ** p < 0.01 was considered indicative of statistically significant differences between the control group and asthma group. Con, sensitized and treated with PBS without drug administration; Asthma, OVA-induced asthma model with untreated; Dexa, OVA-induced asthma model treated with 1 mg/kg of dexamethasone; BC, OVA-induced asthma model treated with 40 mg/kg of BC extract.

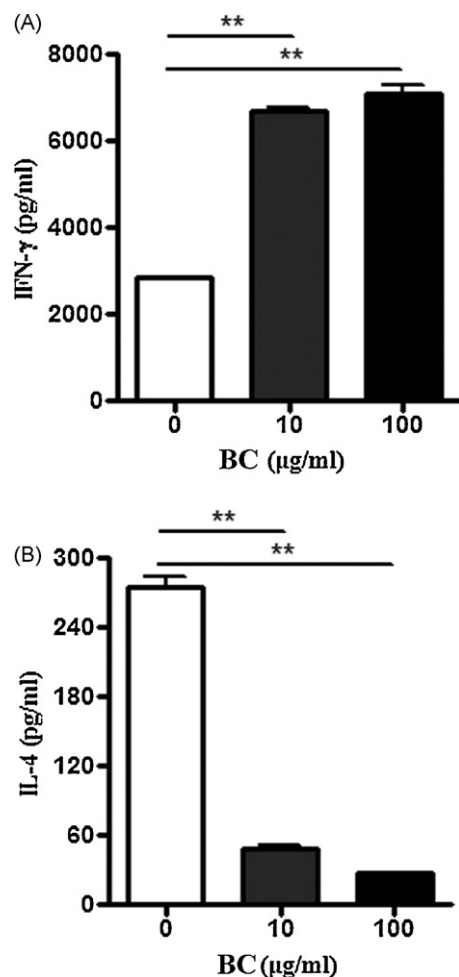


Fig. 5. The production of interferon-gamma (IFN- γ ; A) and interleukin-4 (IL-4; B) on Con A (2 μ g/ml) induced splenocytes. Cells were incubated with 10, 100 μ g/ml of BC extract for 48 h. The production of IFN- γ and IL-4 was analyzed by ELISA as described in Section 2. The data are expressed as the means $\pm$ S.E.M. ** $p < 0.01$ was considered indicative of statistically significant differences between the control group and BC treated groups.

and 2 mM 2-mercaptoethanol and 2 μ g/ml concanavalin A (Con A; Sigma–Aldrich, MO, USA).

2.7. High performance liquid chromatography (HPLC) analysis of BC and *P. nigra* extracts

HPLC analysis was performed on a Waters system (Milford, MA, USA) equipped with a photodiode array detector (Waters 996) running Empower software. A Capcellpak (Shiseido, Tokyo, Japan) C18 column (250 mm $\times$ 4.6 mm, i.d.; 5 μ m) was selected for the chemical fingerprint analysis of BA and *P. nigra* var. *henosis*. The UV data of the effluent from the column ranging from 200 to 400 nm were collected. The flow rate was 0.8 ml/min. The mobile phase comprised 0.1% acetic acid in water (solvent A) and 0.1% acetic acid in methanol (solvent B). The gradient program started with 10% B, followed by a linear gradient to 50% B in 10 min and followed by isocratic elution to 30 min. Then linear gradient to 55% B at 35 min, 100% B at 40 min and followed by isocratic elution to 50 min. The injection volume was 10 μ L.

2.8. DART-TOF-MS analysis

DART (direct analysis in real time) ion source (Ion Sense, Tokyo, Japan) coupled to a JMS-T100TD (AccuTOF-TLC) in the positive

ion mode. The operating conditions of a DART ion source were as follow: needle voltage: 3000 V; perforated and grid electrode potentials: 100 and 100 V, respectively. The TOF-MS was set with a first orifice lens of 5 V, ring lens voltage of 10 V, peak voltage of 500 V, and a detector voltage of 2200 V. Samples introduction were carried out by dipping a glass rod in the solution and placing it between the inlet of MS analyzer and the DART ion source.

2.9. Statistical analysis

Statistical analysis of the data was carried out using the Prism 4 software (GraphicPad Software Inc., CA, USA). Data were presented as mean $\pm$ S.E.M., and statistical analysis was performed by one-way ANOVA for multiple comparisons. Results with $p < 0.05$ were considered statistically significant.

3. Results

3.1. BC extract reduces the infiltration of eosinophils into the airways

To investigate the effect of BC extract in an OVA-induced asthma model, we observed the number of eosinophils in the BAL fluid. The induced infiltration of eosinophils is a main phenomenon in asthma patients. There were a few eosinophils in the control group, but OVA significantly infiltrated the numbers of eosinophils in the asthma group ($p < 0.01$). In contrast, these groups treated with BC extract (10 mg/kg) and Dexa (1 mg/kg) via orogastric gavage markedly inhibited 65.2% and 48.5% OVA-induced infiltration of eosinophils individually (Fig. 1A, $f = 4.119$ and $p = 0.0242$). Furthermore, BC extract and Dexa significantly decreased the eosinophil levels in comparison with the asthma group ($p < 0.01$ and $p < 0.001$) (Fig. 1B, $f = 30.17$ and $p < 0.0001$). These data reveal that BC extract exerted anti-inflammatory effects in an asthma model that is similar to Dexa.

3.2. BC extract inhibits the pathologic progress in lung tissues of asthma model

Infiltration of leukocytes was induced in lung tissue as an index of asthma development. To examine the inhibitory effects of BC extract on the histological features of the OVA-induced asthma model, we stained lung tissues using hematoxylin–eosin staining solution. Although the lung tissues from the OVA-treated group revealed a narrowing of bronchi by infiltration of leukocytes, BC extract significantly reduced the degree of inflammatory cell infiltration ($p < 0.01$) (Fig. 2A). A peribronchial cell count based on a five-point scoring system was performed as described in Section 2. In the asthma group, the inflammation score was significantly increased at 3.3 in comparison with control group at 0.8 ($p < 0.001$). In contrast, BC extract and Dexa group was significantly reduced at 1.2 and 1.9 individually ($p < 0.001$ and $p < 0.05$) (Fig. 2B, $f = 46.02$ and $p < 0.0001$). These data indicate that BC extract reduced the pathological degree in the OVA-induced asthma model.

3.3. BC extract decreases the expression of IL-4 in BAL fluid

Allergic asthma is associated with an enhanced Th 2 lymphocytes response to allergens. IL-4, as represented Th 2 cytokine, plays important roles in the phenomenon of asthma. As BC extract reduced the pathological degree in lung tissues of the OVA-induced asthma model, we investigated whether BC extract decreased the expression of IL-4 in BAL fluid. The results showed that BC extract significantly decreased the OVA-increased protein level of IL-4 in

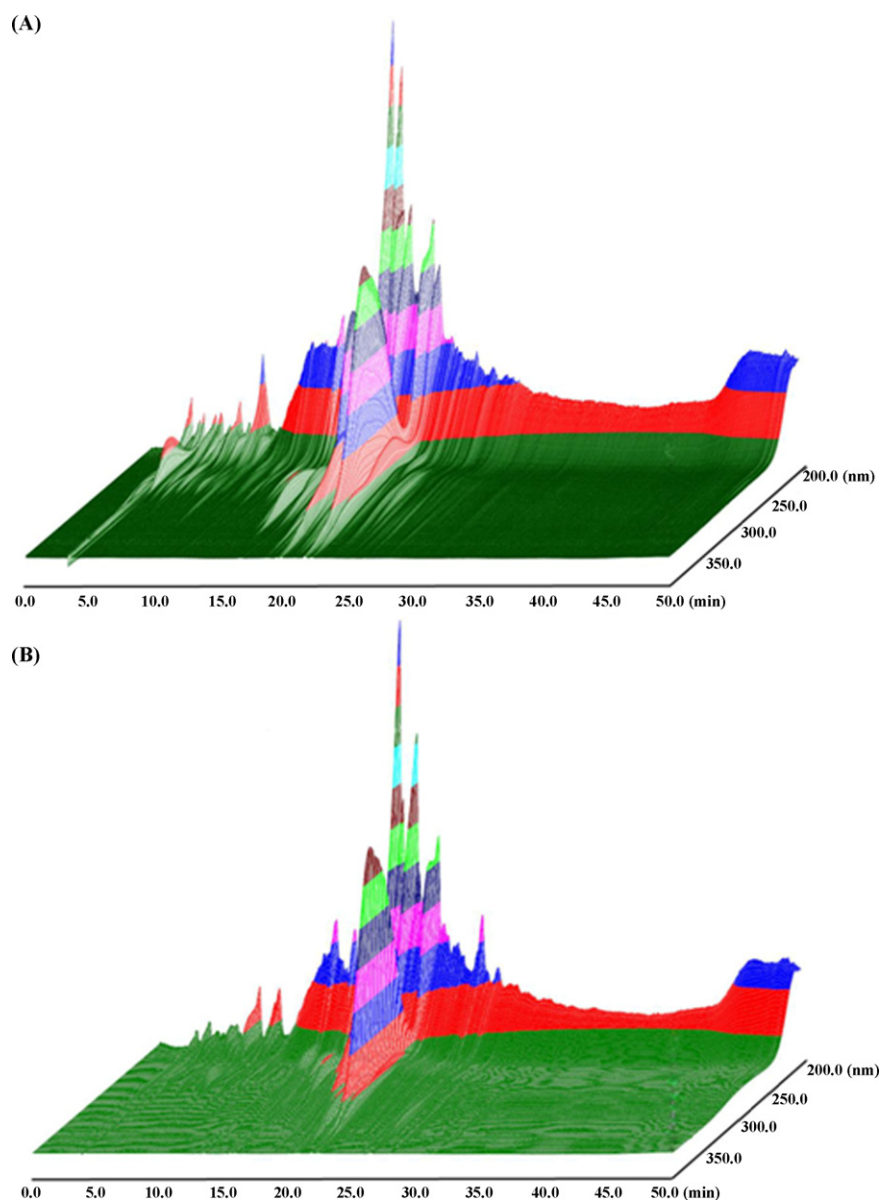


Fig. 6. Three-dimensional (3D) HPLC-PDA fingerprints of the BC extract (A) and *P. nigra* var. *henosis* (B).

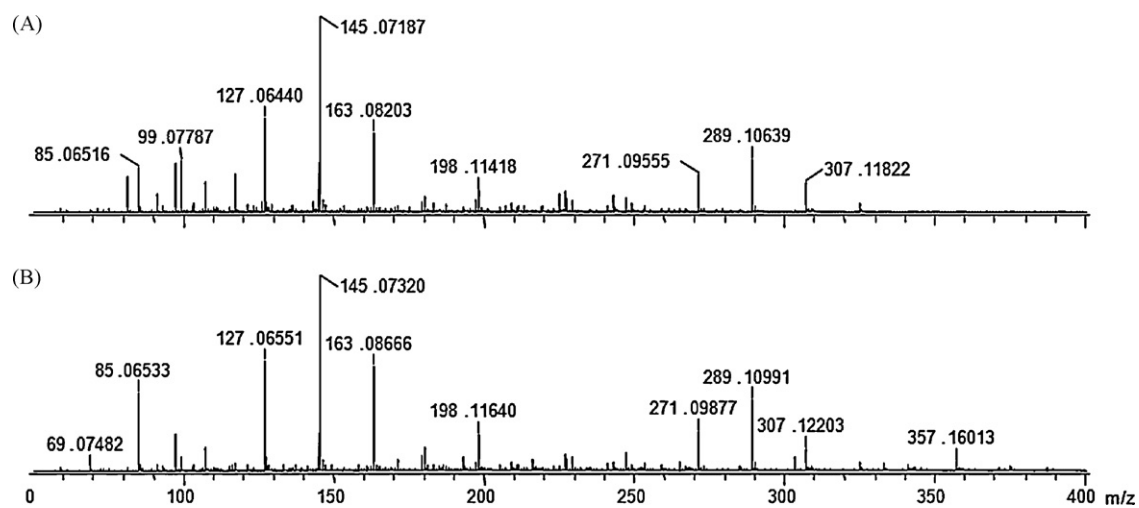


Fig. 7. DART-MS fingerprints of the BC extract (A) and *P. nigra* var. *henosis* (B).

BAL fluid by 55% ($p < 0.001$), which was comparatively more effective than Dexamethasone ($p < 0.05$) (Fig. 3, $f = 10.72$ and $p < 0.0001$).

3.4. BC extract regulates IgE levels in serum

The IgE level in serum is an important factor in allergic asthma and is induced by IL-4 (Pene et al., 1988). Thus, to investigate whether BC extract regulated the levels of total IgE, the concentration of total IgE in serum was measured by ELISA. Although OVA increased total IgE levels in serum, BC extract down-regulated total IgE levels like Dexamethasone, but these results were not significant (Fig. 4, $f = 5.648$ and $p = 0.0071$). The result is correspondence with reduced IL-4 in BAL fluid, previous shown in Fig. 3.

3.5. BC extract up-regulates the IFN- γ and down-regulates IL-4 production from splenocytes in vitro

Since BC extract regulated Th 2 responses (Figs. 1–4), we investigated whether BC extract modulated the Th lymphocyte responses in vitro. Th cells differentiated into two subset; Th 1 lymphocytes or Th 2 lymphocytes. Th 1 and 2 lymphocytes produced IFN- γ and IL-4 cytokines, respectively. Furthermore, IFN- γ and IL-4 cytokine inhibit the growth/differentiation of the opposing subset. Six-week-old Balb/C mice were sacrificed and splenocytes were collected and stimulated with Con A and BC extracts (0, 10 and 100 $\mu\text{g/ml}$) in vitro for 48 h as described in Section 2. The IFN- γ production was significantly increased ($p < 0.01$) and the IL-4 production was significantly decreased ($p < 0.01$) in the supernatants obtained from splenocyte culture that was treated with BC extract (Fig. 5A, $f = 378.8$ and $p = 0.0002$, Fig. 5B, $f = 639.6$ and $p = 0.0001$). These results correspond with the modulation of Th lymphocytes responses.

3.6. HPLC analysis showed the similarity between BC and *P. nigra* extracts

The chemical identity of BC extract was confirmed by direct HPLC chromatogram comparison with the extract of authentic sample of *P. nigra* var. *henosis*. In the chromatogram of BC and *P. nigra* var. *henosis* extracts, the main component showed same retention time of 22 min (Fig. 6). In addition, the entire pattern of chromatogram showed that the major components of the extract of BC and *P. nigra* var. *henosis* extract were identical.

3.7. DART-MS analysis showed the similarity between BC and *P. nigra* extracts

The DART-MS fingerprint of BC extract was directly compared with MS spectrum of the extract from authentic sample of *P. nigra* var. *henosis* locally collected. In the spectra of BC and *P. nigra* var. *henosis* extracts, most of major molecular ion peaks were overlapped (Fig. 7). In addition, the entire pattern of MS spectra showed that two samples from the extract of BC and locally collected *P. nigra* var. *henosis* extracts were almost identical.

4. Discussion and conclusion

Asthma is a chronic disease characterized by reversible airway obstruction and airway inflammation (Hansel and Barnes, 2001). Airway inflammation is one of the important factors in pathologic progress of asthma, and is current target of treatment for suppressing asthma (Walsh, 2006). The number of asthma patients is rapidly increasing and more people received medical treatment for asthma, however most of therapies for asthma are not very effective. In other side, it is increasing that the interest for CAM as asthma therapies (Li and Srivastava, 2006; Li, 2007; Engler et al., 2009).

Thus, new therapeutic medicines from CAM, as adjunct, are needed. The medicine from crude drug has fewer side effects and more efficacy with or without current medicines. Although Chinese herbal medicines have uncertain values in therapies, a number of studies were reported that the effect of herbal medicines in asthma model (Yim et al., in press; Yang et al., 2008; Babayigit et al., 2009).

Dexamethasone is glucocorticoid receptor agonist. Glucocorticoids, including Dexamethasone, are the most effective medicine for asthma (Barnes, 1995). As there were side effects such as no response with steroid in severe diseases (Mjaanes et al., 2006), polarization of Th 2 response (Ramirez, 1998; Wu et al., 1998), however its use is limited. Thus, it has required for medicines that are safer and effective. BC was investigated the anti-inflammatory property by some studies. In recent, *P. nigra*, a main part of BC, was studied that effects on the development of atopic dermatitis and hyperlipidemia (Ham et al., 2009; Qi et al., 2009). In present study, we investigated whether BC extract has anti-inflammatory effect in OVA-induced asthma murine model with focusing on the possibility of BC extract as a therapeutic candidate.

The migration of inflammatory cell into lung is important step in pathogenesis of asthma. In particular, recruitment and activation of eosinophils into the airways of asthma patients is a contributing causative agent in the histopathologies and lung dysfunction (Jacobsen et al., 2007). Thus, it was investigated the number of eosinophils in our OVA-induced asthma model. In asthma group, the number of eosinophil is induced by OVA comparison with control group, however it is decreased by BC extract (Fig. 1A and B). Furthermore, BC extract significantly reduced the infiltration of inflammatory cell into lung tissue, it was correspondence with the results of Fig. 1 (Fig. 2). These results indicated that BC extract has anti-inflammatory effect in the development of asthma.

Th 2 lymphocytes play an important role in allergic disease such as asthma and release IL-4 (Brightling et al., 2002). In particular, the regulation of IL-4 expression has very interesting. IL-4 acts as a growth factor for Th 2 cells, and play a role in the induction of immunoglobulin class switching to immunoglobulin E, which plays a critical role in mediating allergic responses. Further, IFN- γ was a major regulatory cytokine that inhibits the proliferation of Th 2 cells in vitro (Gajewski and Fitch, 1988) and the Th 2 response in vivo (Fernandez-Botran et al., 1988). We therefore measured the IL-4 expression in BAL fluid and IgE in the serum of asthma model. BC extract reduced the induced IL-4 in BAL fluid and IgE in serum by OVA (Figs. 3 and 4). These results suggest that BC extract may regulate the response of Th 2 lymphocytes. It was next investigated whether BC extract modulated Th differentiation into Th 1 and 2 lymphocytes. Th 1 and 2 lymphocytes produce IFN- γ and IL-4 cytokine, respectively. Thus, IFN- γ and IL-4 cytokine mean the differentiation into Th 1 and 2 lymphocytes, respectively. IFN- γ production was significantly increased, while IL-4 production was significantly decreased, after treatment of the culture supernatant of splenocytes with BC extract (Fig. 5). This result suggests that BC extract weaken to differentiate into Th 2 lymphocytes. In another study, IFN- γ mRNA was increased and IL-4 mRNA was decreased in spleen, corresponding to our study (Qi et al., 2009). These results, were shown in Figs. 3 and 5, were suggesting that BC extract might modulate Th subpopulation toward Th 1 response.

The level of IgE is an important factor in allergic disease, such as asthma and is the target for therapy of asthma patient (Humbert and Tonnel, 2005). Mast cells are de-granulated by cross-linking of IgE, this process is important in the development of asthma (Owen, 2007). We observed that the level of IgE is elevated in serum from OVA-induced asthma model. However, the increased level of IgE is markedly decreased by BC extract. This data indicated that BC extract has an inhibition effect on increase of IgE production.

As the authenticity and standardization of the herbal drug material is one of the most important starting point of whole research

project, our study confirmed the authenticity of the material. From both HPLC chromatograms of BC extract (commercial) and *P. nigra* (locally collected and prepared), no significant differences were found from chemical fingerprint pattern (Fig. 6). To confirm the identity of both samples in more detail, we performed another DART-MS analysis for two samples. As shown in Fig. 7, the two spectra showed that the sample we used for the experiment was identical with the authentic sample.

In conclusion, we discovered that BC extract reduced the infiltration of inflammatory cell into lung tissues, the expression of IL-4 in BAL fluid and IgE production in serum from OVA-induced asthma model. Furthermore, BC extract up-regulates the IFN- γ and down-regulates IL-4 production in the supernatants of splenocytes culture, suggesting that BC extract enhanced Th differentiation into Th 1 lymphocytes in splenocytes culture. Together all results, BC extract reduces OVA-induced airway inflammation and Th 2 response in mice. Thus, our study was suggesting that BC extract can be a therapeutic candidate for allergic airway disease, including asthma.

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